1.1. $g(x,y) = x^2 - y^2 - 2x + 4y + 6$

$g_x = 2x - 2 = 0 \quad \rightarrow \quad x = 1$

$g_y = -2y + 4 = 0 \quad \rightarrow \quad y = 2$

Critical Point: (1, 2)

Second-Order Derivatives:

$g_{xx} = 2$

$g_{yy} = -2$

$g_{xy} = 0$

Hessian Determinant (D):

$D = (g_{xx})(g_{yy}) - (g_{xy})^2$

$D = (2)(-2) - (0)^2 = -4$

Conclusion:

Since $D < 0$, the point (1, 2) is a SADDLE POINT.

1.2. $g(x,y) = 2x^2 - 2xy + y^2 + 5x - 3y$

$g_x = 4x - 2y + 5 = 0$
$g_y = -2x + 2y - 3 = 0$

Solving the system:
$(4x - 2y + 5) + (-2x + 2y - 3) = 0$
$2x + 2 = 0 \quad \rightarrow \quad x = -1$
Substitute into g_y:
$-2(-1) + 2y - 3 = 0$
$2y - 1 = 0 \quad \rightarrow \quad y = 0.5$
Critical Point: $(-1, 0.5)$

Second-Order Derivatives:
$g_{xx} = 4$
$g_{yy} = 2$
$g_{xy} = -2$

Hessian Determinant (D):
$D = (4)(2) - (-2)^2$
$D = 8 - 4 = 4$

Conclusion:
Since $D > 0$ and $g_{xx} > 0$, the point $(-1, 0.5)$ is a LOCAL MINIMUM.

1.3. $u(x,y) = 4x + 2y - x^2 + xy - y^2$

$u_x = 4 - 2x + y = 0$
$u_y = 2 + x - 2y = 0$

Solving the system:
From u_x: $y = 2x - 4$
Substitute into u_y:
$2 + x - 2(2x - 4) = 0$
$2 + x - 4x + 8 = 0$
$10 - 3x = 0 \quad \rightarrow \quad x = 10/3 \ (\sim 3.33)$
$y = 2(10/3) - 4 = 8/3 \ (\sim 2.67)$
Critical Point: (3.33, 2.67)

Second-Order Derivatives:
$u_{xx} = -2$
$u_{yy} = -2$
$u_{xy} = 1$

Hessian Determinant (D):
$D = (-2)(-2) - (1)^2 = 3$

Conclusion:
Since $D > 0$ and $u_{xx} < 0$, the point (3.33, 2.67) is a LOCAL MAXIMUM.

2.1. $f(x,y) = 3x^2 + 4y^2 - xy$ subject to $2x + y = 21$

Method: Substitution

$y = 21 - 2x$

Substitute into f(x,y):

$f(x) = 3x^2 + 4(21 - 2x)^2 - x(21 - 2x)$

$f(x) = 3x^2 + 4(441 - 84x + 4x^2) - 21x + 2x^2$

$f(x) = 3x^2 + 1764 - 336x + 16x^2 - 21x + 2x^2$

$f(x) = 21x^2 - 357x + 1764$

Optimization:

$f'(x) = 42x - 357 = 0$

$42x = 357 \quad \rightarrow \quad x = 8.5$

$y = 21 - 2(8.5) = 4$

Second Derivative:

$f''(x) = 42$

Since $f''(x) > 0$, the function has a MINIMUM at this point.

Calculation of f(8.5, 4):

$f = 3(8.5)^2 + 4(4)^2 - (8.5)(4)$

$f = 216.75 + 64 - 34 = 246.75$

2.2. $f(x,y) = 5x^2 + 6y^2 - xy$ subject to $x + 2y = 24$

Method: Substitution

$x = 24 - 2y$

Substitute into f(x,y):

$f(y) = 5(24 - 2y)^2 + 6y^2 - y(24 - 2y)$

$f(y) = 5(576 - 96y + 4y^2) + 6y^2 - 24y + 2y^2$

$f(y) = 2880 - 480y + 20y^2 + 6y^2 - 24y + 2y^2$

$f(y) = 28y^2 - 504y + 2880$

Optimization:

$f'(y) = 56y - 504 = 0$

$56y = 504 \quad \rightarrow \quad y = 9$

$x = 24 - 2(9) = 6$

Second Derivative:

$f''(y) = 56$

Since $f''(y) > 0$, the function has a MINIMUM at this point.

Calculation of f(6, 9):

$f = 5(6)^2 + 6(9)^2 - (6)(9)$

$f = 180 + 486 - 54 = 612$

3.1. $f(x,y) = x^2 + 24xy + 8y^2$ subject to $x^2 + y^2 = 25$

Method: Lagrange Multipliers
$L = x^2 + 24xy + 8y^2 - \lambda(x^2 + y^2 - 25)$

First-Order Conditions:
$L_x = 2x + 24y - 2\lambda x = 0 \rightarrow x(1 - \lambda) = -12y \quad (1)$
$L_y = 24x + 16y - 2\lambda y = 0 \rightarrow 24x = y(2\lambda - 16) \quad (2)$

From (1): $x/y = -12 / (1 - \lambda)$
From (2): $x/y = (2\lambda - 16) / 24$

Equating the two:
$-12 / (1 - \lambda) = (2\lambda - 16) / 24$
$-288 = (1 - \lambda)(2\lambda - 16)$
$-288 = 2\lambda - 16 - 2\lambda^2 + 16\lambda$
$2\lambda^2 - 18\lambda - 272 = 0$
$\lambda^2 - 9\lambda - 136 = 0$
$(\lambda - 17)(\lambda + 8) = 0$
Roots: $\lambda = 17, \lambda = -8$

Case 1: $\lambda = 17$
$x/y = -12 / (1 - 17) = 3/4 \rightarrow x = 0.75y$
Constraint: $(0.75y)^2 + y^2 = 25$
$0.5625y^2 + y^2 = 25$
$1.5625y^2 = 25 \rightarrow y^2 = 16 \rightarrow y = \pm 4, x = \pm 3$
$f(3, 4) = 425$ (MAXIMUM)

Case 2: $\lambda = -8$
$x/y = -12 / (1 + 8) = -4/3 \rightarrow x = -1.33y$
Constraint: $(-4/3\ y)^2 + y^2 = 25$
$16/9\ y^2 + y^2 = 25$
$25/9\ y^2 = 25 \rightarrow y^2 = 9 \rightarrow y = \pm 3, x = \mp 4$
$f(4, -3) = -200$ (MINIMUM)

3.2. $f(x,y) = x^2 - 10y^2$ subject to $x - y = 18$

Method: Lagrange Multipliers
$L = x^2 - 10y^2 - \lambda(x - y - 18)$

First-Order Conditions:
$L_x = 2x - \lambda = 0 \quad \rightarrow \quad \lambda = 2x$
$L_y = -20y + \lambda = 0 \quad \rightarrow \quad \lambda = 20y$
Therefore: $2x = 20y \quad \rightarrow \quad x = 10y$

Constraint:
$x - y = 18$
$10y - y = 18$
$9y = 18 \quad \rightarrow \quad y = 2$
$x = 20$

Bordered Hessian for Second-Order Condition:
Constraint $g(x,y) = x - y - 18$
$g_x = 1, g_y = -1$
$L_{xx} = 2, L_{yy} = -20, L_{xy} = 0$

$$|H| = \begin{vmatrix} 0 & g_x & g_y \\ g_x & L_{xx} & L_{xy} \\ g_y & L_{xy} & L_{yy} \end{vmatrix} = \begin{vmatrix} 0 & 1 & -1 \\ 1 & 2 & 0 \\ -1 & 0 & -20 \end{vmatrix}$$

Determinant:
$|H| = 0 - 1(-20 - 0) + (-1)(0 - (-2))$
$|H| = 20 - 2 = 18$
Since $|H| > 0$, the point (20, 2) is a MAXIMUM.

Calculation of Value:
$f(20, 2) = (20)^2 - 10(2)^2 = 400 - 40 = 360$